



How many chemicals in beer?

German chemists identified over 7,700 different chemical formulas in beers.
<https://digit.in/sep21-21>



Oil eating bacteria

Researchers say arctic marine bacteria can biodegrade fossil fuels
<https://digit.in/sep21-22>

Source: MIT News



A researcher lifting a sheet of the kombucha inspired material

compounds can be used to do things like track pollutants in the atmosphere. The reason that it is such a significant development because as an MIT student, who is a lead author of the paper that talks about this discovery, Tzu-Chieh Tang, said in a statement published on MIT News, “We think this is a good system that is very cheap and very easy to make in very large quantities... It’s at least a thousand times more material than the E. coli system.”

Pollutant detectors made using tea fermentation techniques are not the only applications of living materials. Things like building materials and bioelectronics are being researched upon, too. The researchers at the University of Chicago have recently found a way in which they have found a way to make biological tissues communicate effectively with electronics, which are complex lifeless machines programmed to work in a specific way. What this does is that it adds another dimension to the workings of devices, which can now adapt in real-time to alter their functionality by reacting to stimuli, which we humans can sense through the

Source: CU Boulder Today



A brick made from living materials in its mould

cells and tissues in our body. When it comes to building materials, the researchers at the University of Colorado, Boulder, have fabricated or tapped into cyanobacteria belonging to the genus *Synechococcus*. As a part of their research, the minds at CU Boulder have been able to make the bacteria help in the production of bricks.

They found that under certain specific conditions, the bacteria under study could absorb carbon dioxide from the atmosphere, which helped them grow and transform into calcium carbonate. The research is not yet at a stage when large scale manufacturing of building materials can begin. The team is now looking into approaches by which the process can be scaled.

In January of 2020, a team led by MIT Media Lab Associate Professor Neri Oxman, and consisting of people from MIT itself, and Harvard University’s Wyss Institute and Dana-Farber Cancer Institute, developed something they call as “hybrid living materials, or HLMs”. During their research, the team found a way to integrate chemical signals into the resins they use for printing materials from the multi-material 3D printers. After this, biologically engineered bacteria were introduced onto the surface of the printed material by spraying it on, infused in the hydrogel.

When sprayed, changes known as enzymatic reactions were triggered in these microorganisms. This was observed as a colour change on the surface of the material, which the researchers claim, shows the formation of different biosynthetic products created by the bacteria introduced onto the surface of the 3D printed object. Interestingly, these new HLMs can be programmed by the developers of these materials to make the bacteria behave in a certain way. This can

come in handy in, say, applications like medicine administration, which uses a 3D printed material, which, when triggered by a particular situation inside your body, can cause the microorganism on it to release certain chemicals to make sure everything is back to normal. It will be like a hospital inside your body that is installed and gets triggered when things go wrong in the systemic functioning of the organs inside. The possibilities are immense. Another such application of the living materials can be in the world of sports where the clothing of, say, a swimmer can change its stiffness to cling on to the body and cling on when the swimmer is in the water. When the swimmer comes out,



Source: MIT News

3D printed HLMs may be the gateway to new wearable or medicinal tech

the swimsuit loses out again after the water has dried off. Again, the only problem that remains is that of the production scales. Doing things in a lab is one thing, and producing it for the people in a cost-effective manufacturing process is another. Matching the cost and effectiveness of the conventional processes is a hurdle that these researchers have to overcome to bring living materials to the world around us. Materials may be given novel properties by fusing them with microorganisms that react to environmental inputs, but the development does raise a question. So far, our smartphones could only be infected by viruses of the digital kind, but with bioelectronics, users will have to be extra careful about their touchscreens, because they can potentially be infected by viruses of the biological kind as well. Thank you, scientists. **d**